

Tofa Kimani Mwangi

Data & Software Engineering | Backend Systems & Data Infrastructure | Big Data

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PROFESSIONAL SUMMARY

Applied Computing graduate and ALX-certified Data Engineer with hands-on experience designing end-to-end ETL pipelines, star-schema data warehouses, and automated analytics workflows. Proficient in Python, SQL, PySpark, Apache Airflow, PostgreSQL, and containerization using Docker. Proven track record of optimizing data integrity and processing large-scale telecom and healthcare datasets for operational reporting.

TECHNICAL SKILLS

- Programming Languages:** Python, JavaScript, TypeScript, SQL, Java, Bash.
- Software & API Engineering:** Node.js, Express, Next.js, RESTful APIs, Role-Based Access Control (RBAC), Springboot, Flyway.
- Big Data & Data Engineering:** PySpark, Apache Spark, Apache Airflow, ETL/ELT Pipelines, Data Warehousing, Dimensional Modeling (Star Schema), BigQuery, Pandas, NumPy.
- Databases:** PostgreSQL, MySQL, MongoDB, Database Indexing, Query Optimization, Schema Normalization.
- Data Quality & Governance:** Pipeline Testing & Quality Gates, Schema Enforcement, Column-Level Minimization, Sensitive Data Protection, Lineage & Auditability.
- DevOps & Infrastructure:** Docker, Docker Compose, Linux, Git/GitHub, CI/CD Basics.
- System Design & Architecture:** System Design, RESTful APIs, Monolith Architecture, Database Indexing & Query Optimization.
- Soft Skills:** Critical Thinking, Cross-Functional Collaboration, Problem-Solving & Troubleshooting, Technical Communication, Attention to Detail, Time Management, Stakeholder Alignment, Adaptability, Agile Methodology

WORK EXPERIENCE

Corporate Governance Data Analyst – Virtual Assistant

Minerva Analytics (Remote)

January 2026 – April 2026

- Audited public company filings, annual reports, and proxy statements to log board structure changes, executive leadership shifts, and committee appointments.
- Captured and verified financial metrics, director independence data, and board governance updates with high accuracy across global markets.

IT Intern

Social Health Authority, Nanyuki, Kenya

September 2024 – December 2024

- Cleaned and prepared large-scale datasets using Excel and internal diagnostic tools, reducing reporting errors and optimizing data integrity.
- Conducted operational data analysis to identify bottlenecks and collaborated with cross-functional teams to redesign IT workflows, accelerating report generation.
- Resolved technical hardware, software, and network issues across computer systems to ensure high uptime for critical user applications.

PROJECTS

Zuri Mobile Customer Churn ETL Pipeline

PySpark, Apache Airflow, PostgreSQL, Docker Compose

June 2026

- Engineered a production-style telecom churn pipeline processing 10,000 records using PySpark for data validation, deduplication, and standardization.
 - Designed a PostgreSQL star-schema data warehouse to optimize dimensional modeling and analytics performance.
 - Orchestrated automated, repeatable staging-to-live data workflows using Apache Airflow and Docker Compose.
- github.com/TofaKM/zuri-mobile-churn-pipeline

Hospital Data ETL Pipeline

Docker, Python, SQL, ETL Architecture

June 2026

- Developed a containerized ETL pipeline in Docker to automate healthcare data ingestion, cleaning, and transformation.
- Structured end-to-end data workflows generating high-precision datasets for operational analysis and clinical reporting.
- Automated routine transformations to allow non-technical stakeholders direct access to accurate business insights.
github.com/TofaKM/etl_hospital_pipeline

Chap Mall E-Commerce Platform

Next.js, Node.js, PostgreSQL, RESTful APIs

July 2025

- Architected a multi-tier e-commerce platform featuring Role-Based Access Control (RBAC) across customer, vendor, and admin roles.
- Engineered RESTful APIs and optimized PostgreSQL database queries to handle complex relational data with high data integrity.
- Built inventory tracking logic for live availability updates and implemented secure authentication to lower page latency.
github.com/TofaKM/Advanced-E-commerce-Website

CERTIFICATIONS

- **Data Engineer Certificate** – ALX Africa (Aug 2026) — ETL/ELT Pipelines, Data Warehousing, Airflow, Spark, Dimensional Modeling, SQL/Python Automation.
- **Python for Data Science** – ALX Africa (Apr 2026) — Pandas, NumPy, SciPy, Automated Data Extraction, Statistical Hypothesis Testing.
- **Data Analytics Certificate** – ALX Africa (Sep 2025) — MySQL Querying, Power BI Dashboarding, Data Visualization, Professional Reporting, Excel.
- **Introduction to Machine Learning** – Cisco & IBM (Feb 2025) — Supervised/Unsupervised Learning, Model Evaluation, Predictive Analytics (Python).
- **Data Analytics** – Cisco Academy (Aug 2024) — Data Lifecycle Management, Data Cleaning, Advanced Excel, SQL Analysis.

EDUCATION

B.Sc. Applied Computing (Computer Science)

KCA University, Nairobi, Kenya

September 2020 – November 2025

- Core Coursework: Data Structures & Algorithms (DSA), Systems Analysis & Design, Database Management Systems (DBMS), Relational Database Normalization, SDLC.
- Capstone Project: Engineered a full-stack e-commerce application using JavaScript, Node.js, React, and MySQL featuring optimized relational schemas and Role-Based Access Control (RBAC).

High School Certificate

Ndururumo High School, Nyahururu, Kenya

January 2016 – November 2019

- Achieved Mean Grade B+ (71 points), focusing on Mathematics, Physics, and Computer Studies.
- Served as Academic Secretary, leading peer tutoring initiatives and managing academic schedules.

INTERESTS

Professional Interests & Research: System Architecture, AI & ML, Tech News, Cloud Data Architecture, Real-Time Data Streaming, Distributed Systems & Microservices

Activities & Hobbies: Football, Badminton, Outdoor Activities, Gardening, Animations

REFERENCES

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- **Francis Ngugi** — *Software Engineer*
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